SUYASH SINGH

(+91) 8604960115 | suyashs787@gmail.com | linkedin.com/in/suyashin | github.com/anothercodingguy

EDUCATION

Manipal Institute of Technology Expected Graduation: 2027
B.Tech in Computer Science Engineering (Data Science) Bengaluru, India

- **CGPA**: 8.51/10 | Strong foundation in CS fundamentals, Data Structures & Algorithms; full-stack development.

TECHNICAL SKILLS

Core Fundamentals: Data Structures & Algorithms, Object-Oriented Design (OOD), System Design, Operating Systems
Languages: Java, C++, Python, TypeScript, JavaScript (ES6+), SQL
Backend & Cloud: Node.js, Express.js, FastAPI, REST APIs, Microservices, Docker, Kubernetes, NATS, KEDA, AWS, GCP
Observability & Data: Prometheus, Grafana, Redis, PostgreSQL, MongoDB, Qdrant Vector DB, Upstash
ML / AI: PyTorch, LLM Pipelines, RAG Frameworks, Machine Unlearning Pipelines
Competitive Programming: LeetCode (200+ Problems Solved) | Codeforces **Pupil** (1224 Max Rating) | CodeChef **3**⋆ (Problem Solving & Python)

TECHNICAL PROJECTS

PathFlow *"Strava for AI Agents"*
Next.js 15, TypeScript, Tailwind CSS, React Flow, Prisma, OpenTelemetry, Python

- **System Architecture**: Architected an OpenTelemetry-compatible observability platform to track execution paths, token velocity (t/s), context volume, and API compute costs across autonomous agent fleets.
- **Interactive Visualization**: Built an interactive DAG visualizer with React Flow to inspect multi-step agent execution trees, sub-span latencies, and tool-routing decisions in real-time.
- **Benchmarking & Telemetry**: Engineered a multi-factor skill benchmarking engine ranking agents by micro-segment (Score = Success Rate × Efficiency × Novelty × Difficulty) and a lightweight Python SDK (`@pf.trace`) streaming spans to the ingestion API.

Semantic LLM Gateway & Routing Proxy
FastAPI, Qdrant, Redis, Hugging Face, Groq, Ollama

- **System Design**: Built a production-grade, AI-integrated proxy with Qdrant-backed semantic caching, drastically reducing redundant LLM API calls and achieving cache-hit latencies **under 50ms**.
- **Complex Routing**: Implemented dynamic intent routing to classify prompt complexity, intelligently distributing concurrent loads across high-performance and lightweight models to optimize resource utilization.
- **Fault Tolerance**: Engineered a robust circuit-breaker fallback mechanism, maintaining **zero-downtime** reliability and high availability during upstream rate limits or network outages.

ReachInbox *Full-Stack Email Scheduler*
TypeScript, Next.js, Node.js, Redis

- **Scalable Architecture**: Architected and deployed a highly concurrent email scheduling system using Next.js and Redis, ensuring reliable asynchronous task execution across distributed queues.
- **Full-Stack Engineering**: Designed responsive interfaces and optimized backend data pipelines, successfully demonstrating the versatility required to solve problems across the entire software stack.

Self-Erasing Neural Networks (SENNs) *Research Publication, ICDDS 2025*
Python, PyTorch, Data Structures & Algorithms

- **Experimental Systems**: Co-authored a peer-reviewed paper designing a complex algorithmic framework for **GDPR-compliant machine unlearning**, accepted at the ICDDS 2025 international conference.
- **Data Analysis & Visualization**: Developed rigorous diagnostic pipelines to evaluate weight shifts and per-class accuracy trade-offs, visualizing performance metrics across gradient-based data erasure models.

WORK EXPERIENCE

Stealth Startup Dec. 2025 – May 2026
AI Intern Bengaluru, India

- Architected and deployed scalable REST APIs and distributed inference pipelines on **AWS**, engineering highly concurrent state machine logic and multi-turn session management.
- Engineered intelligent routing logic and resilient backend services, optimizing concurrent request handling and backend fault tolerance.

IEEE Computer Society Bangalore Chapter Apr. 2025 – Sept. 2025
R&D Intern Bengaluru, India

- Evaluated complex distributed architectures and AI research to build production-ready reference implementations.
- Analyzed system performance metrics and log data to validate architectural reliability across distributed nodes.

POSITIONS OF RESPONSIBILITY

Manipal Bengaluru Open-Source Community (MBOSC) 2024 – 2025
Project Head

- Mentored **200+ student developers** on system architecture, Git workflows, and open-source contributions, leading code reviews and encouraging software design best practices.

Codex (Competitive programming club) 2025
Project Head